



EMBEDDED SYSTEMS PROJECT

# Arduino-Based Closed-Loop Conveyor System

Speed & Direction Control of a DC Motor-Driven Conveyor using a Potentiometer, Arduino Microcontroller, and Real-Time LCD Feedback

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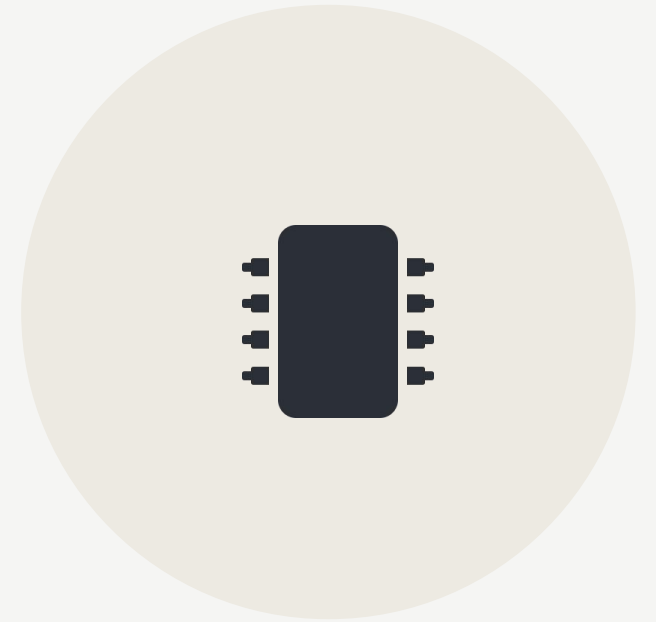
Presented by

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June Batch

# Introduction

- Modern industries rely on conveyor systems for efficient, continuous material handling.
- This project develops a low-cost, closed-loop conveyor controller built around an Arduino microcontroller.
- A potentiometer allows the operator to vary DC motor speed smoothly from 0% to 100%.
- A 16x2 LCD gives real-time visibility into speed, direction, and system status.
- The design illustrates how embedded systems bring precision and monitoring to industrial automation at minimal cost.



# Project Objectives



## Variable Speed Control

Continuously adjust DC motor speed from 0% to 100% using a potentiometer and PWM.



## Direction Control

Enable forward and reverse operation of the conveyor motor as required.



## Real-Time Monitoring

Display speed, direction, and status live on a 16x2 LCD for the operator.

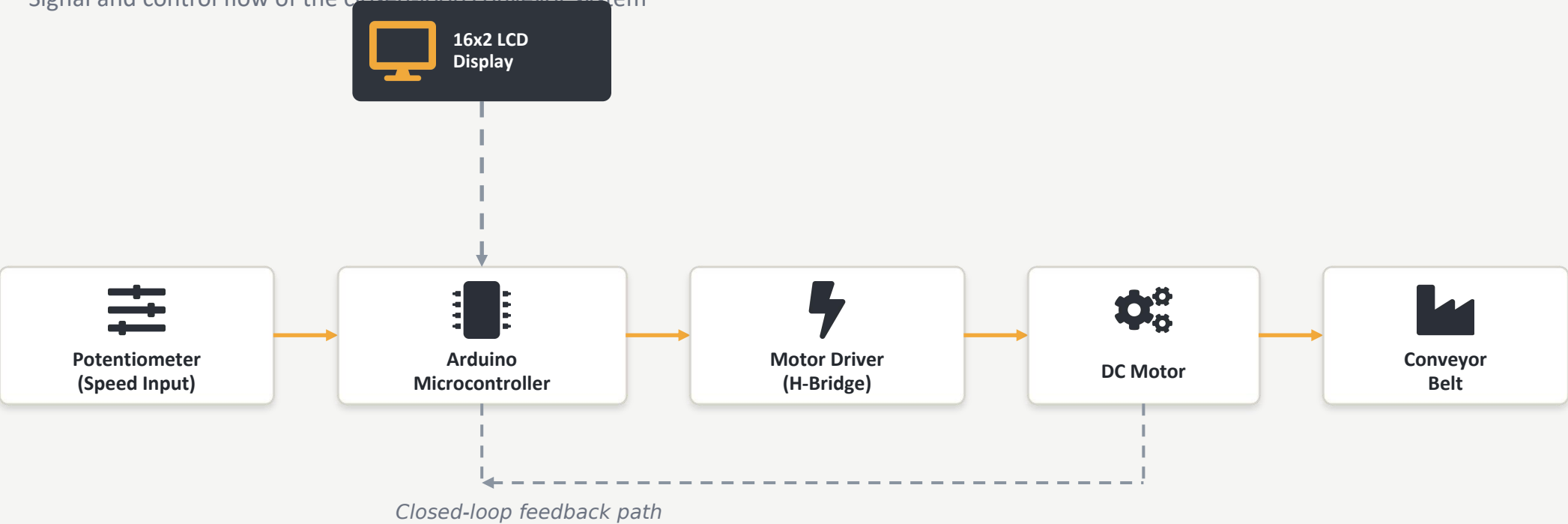


## Closed-Loop Operation

Continuously read feedback and adjust output for stable, reliable performance.

# System Architecture

Signal and control flow of the closed-loop conveyor system





**Input**  
Operator sets desired speed via the potentiometer.



**Processing**  
Arduino converts the reading into a PWM control signal.



**Output**  
Motor drives the conveyor while the LCD reports live status.

# Hardware Components



## Arduino Board

Central controller — reads potentiometer, runs PWM, drives LCD & motor driver.



## Potentiometer

Variable resistor for continuous 0%-100% speed input.



## Motor Driver (H-Bridge)

Drives the DC motor and enables forward / reverse direction control.



## DC Motor

Drives the conveyor belt mechanism at the commanded speed.



## 16x2 LCD Display

Shows real-time speed %, direction, and operating status.

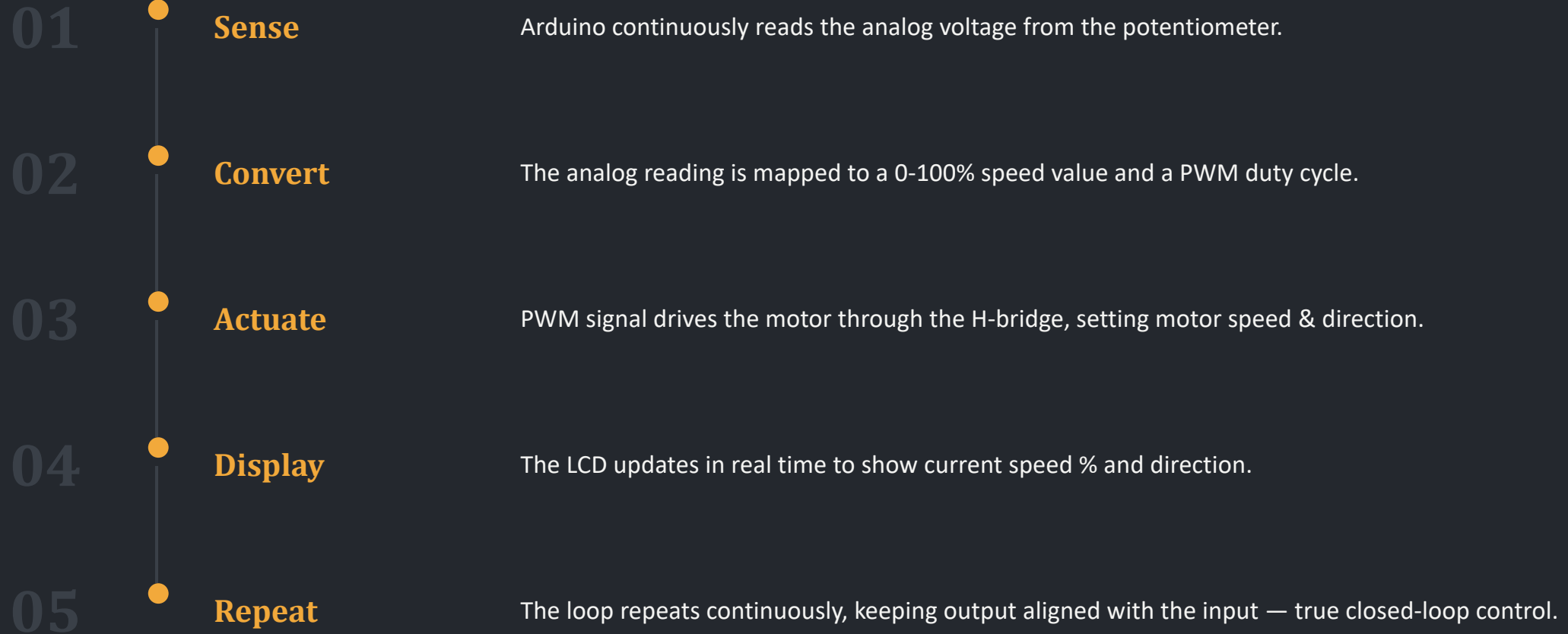


## Conveyor Mechanism

Belt and roller assembly that transports material.

# Working Principle

Closed-loop control cycle



# Key Features

- ✓ Smooth, continuous speed control from 0% to 100%
- ✓ Forward and reverse direction control
- ✓ Real-time parameter display on 16x2 LCD
- ✓ Closed-loop design for improved accuracy and monitoring
- ✓ Simple, low-cost, and easily reproducible hardware
- ✓ Scalable concept applicable to real industrial automation

**0-100%**

Continuous Speed Range

**16x2**

LCD Real-Time Display

**2-Way**

Motor Direction Control

# Applications



## Manufacturing Lines

Material transport and sorting on production floors.



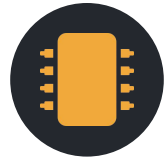
## Packaging Systems

Controlled movement of items through packing stations.



## Warehousing & Logistics

Speed-adjustable conveyors for load handling and dispatch.



## Automation Education

Teaching tool for embedded control and PWM concepts.

# Advantages



Low-cost components keep the build affordable



Real-time feedback improves monitoring and control



Simple circuit — easy to build, test, and maintain



Closed-loop design increases reliability and accuracy



Smooth analog speed control via potentiometer



Concept scales to larger industrial conveyor systems

# Conclusion

The Arduino-Based Closed-Loop Conveyor System successfully demonstrates how embedded-system concepts — sensing, PWM control, and real-time display — combine to deliver precise, flexible, and low-cost conveyor automation. It provides a strong practical foundation for scaling embedded control techniques into real industrial automation applications.

- ✓ Reliable closed-loop speed & direction control
- ✓ Real-time operator feedback via LCD
- ✓ Simple, low-cost, industry-relevant design

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## Thank You

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